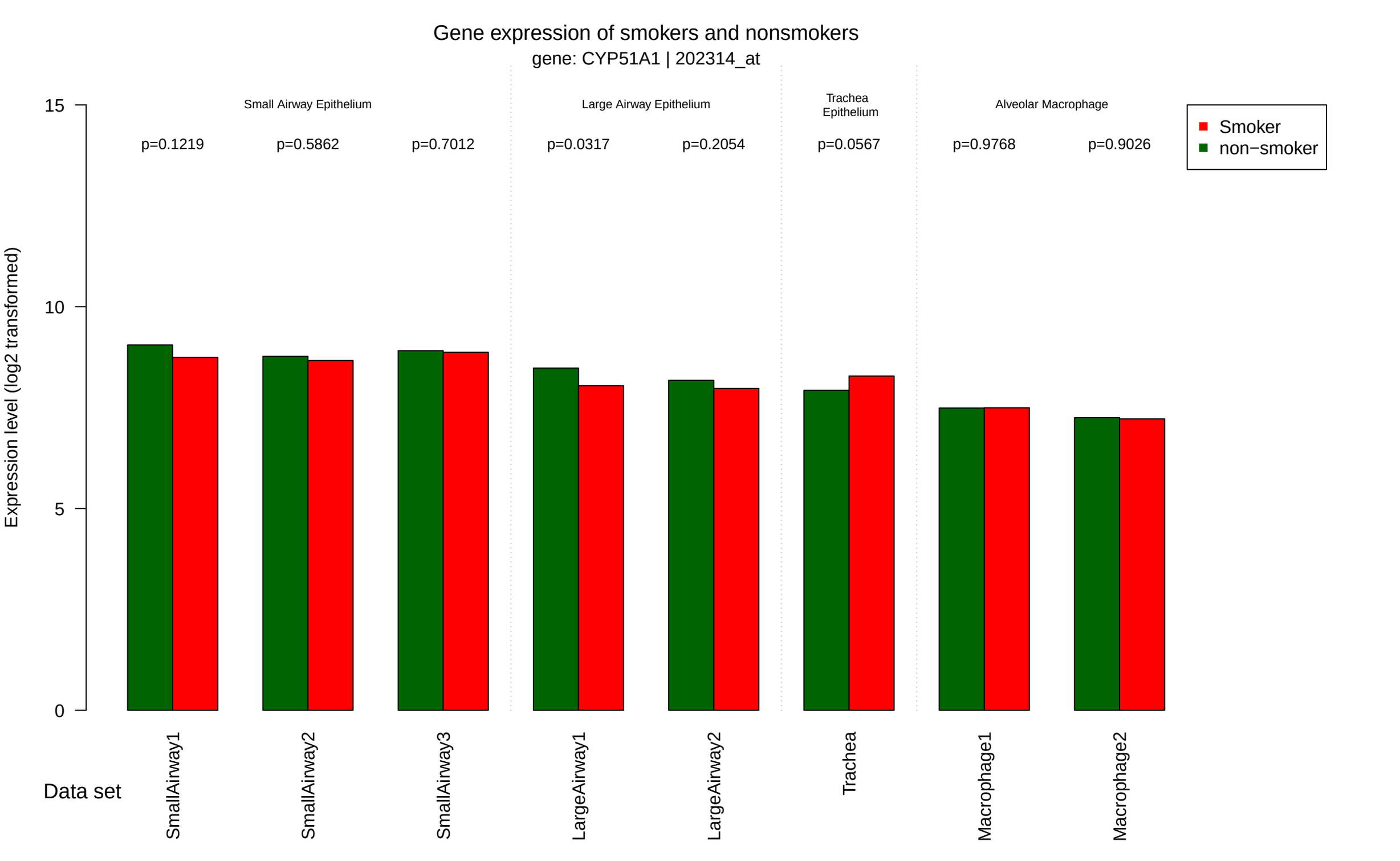
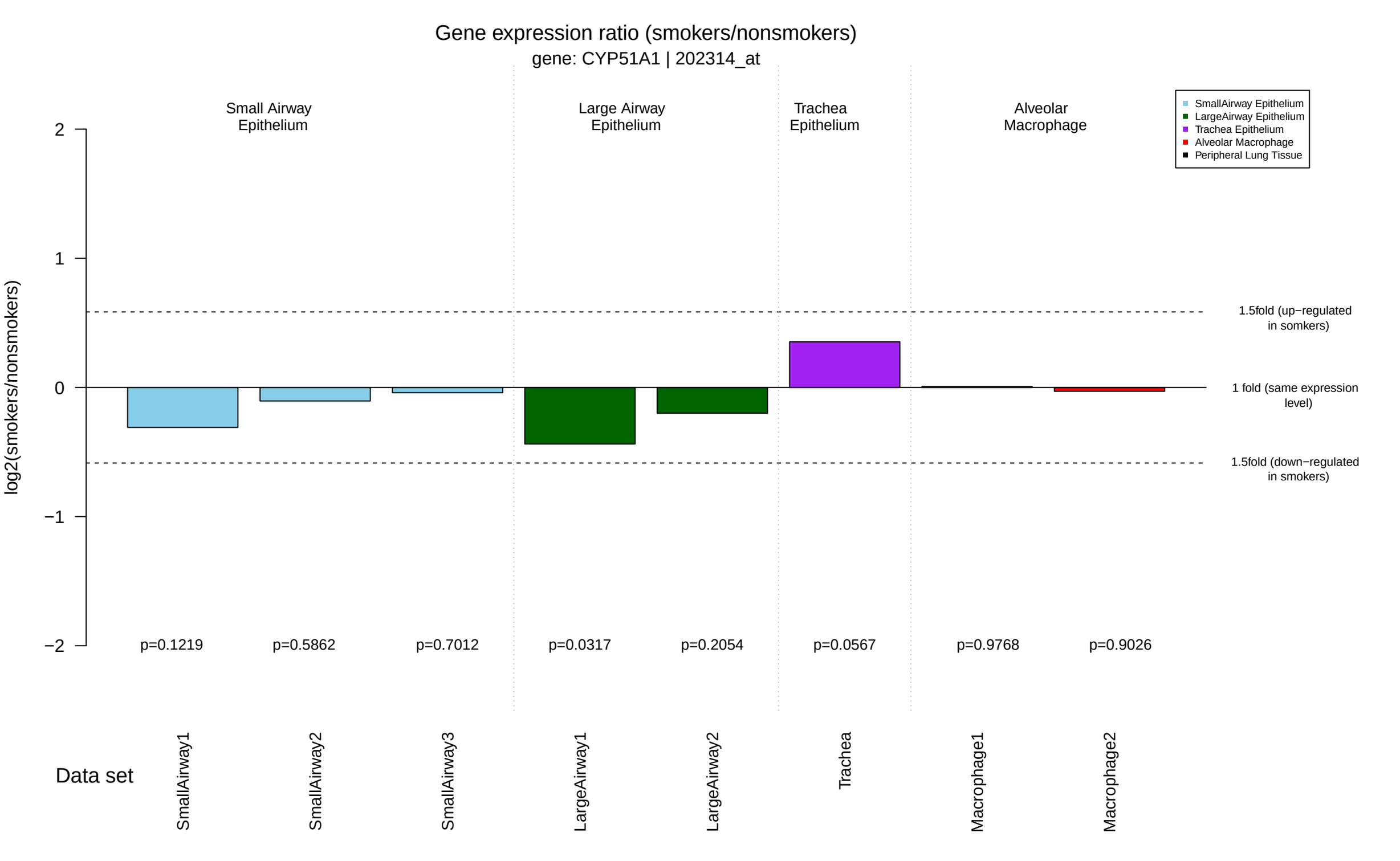
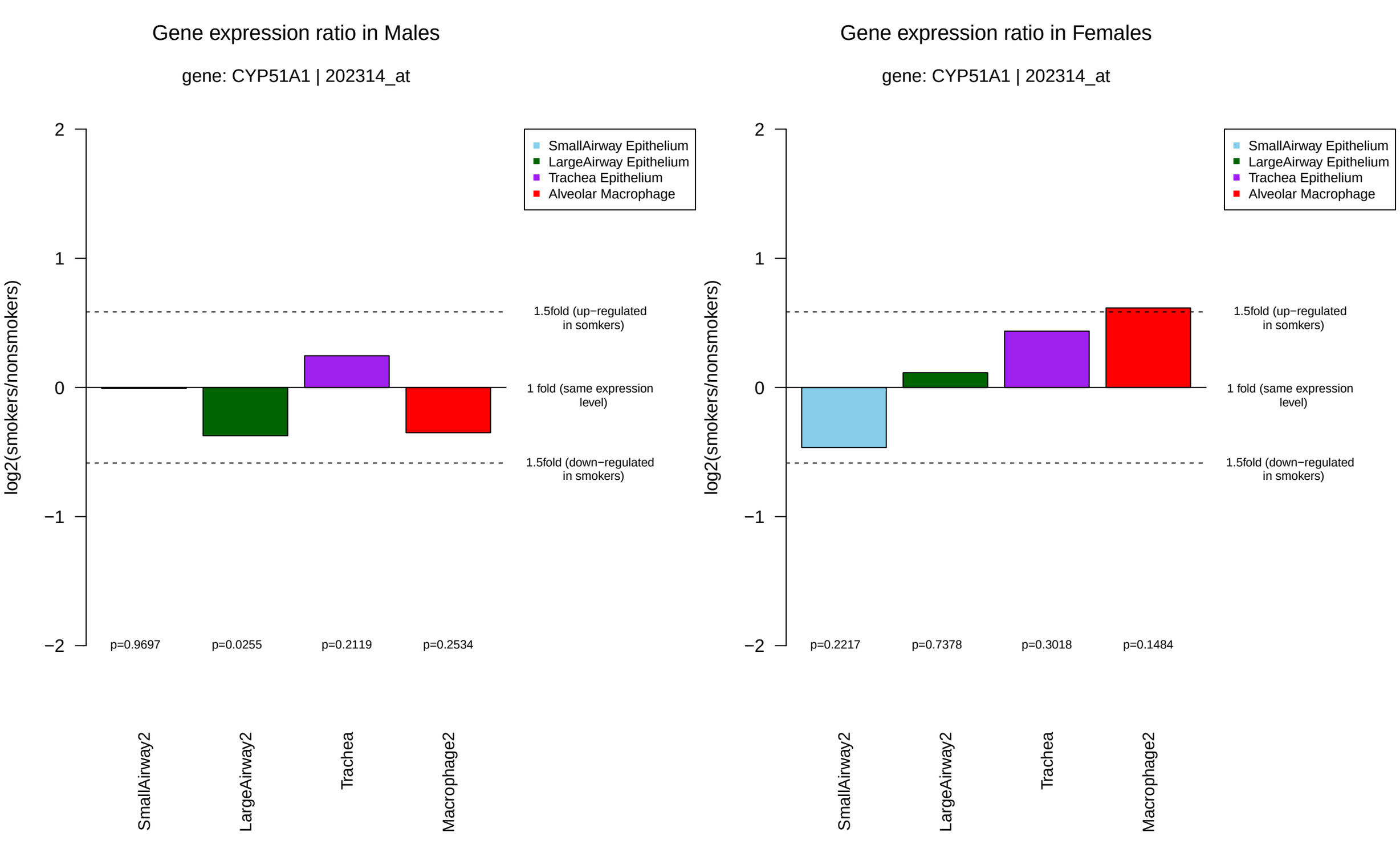
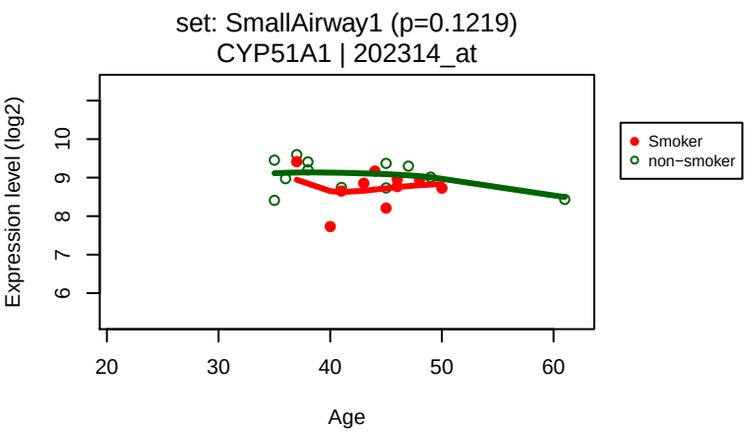
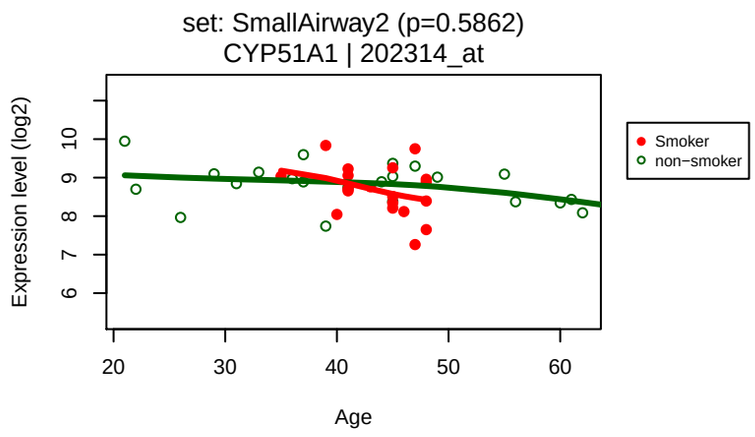
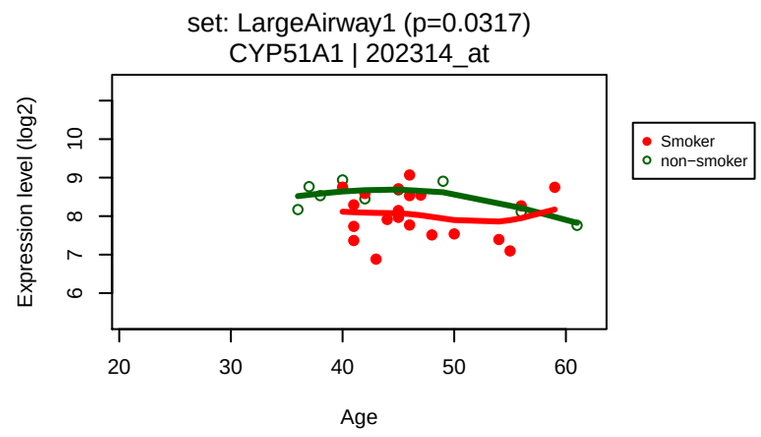

Gene expression of smokers and nonsmokers
gene: CYP51A1 | 202314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1219
p=0.5862
p=0.7012
p=0.0317
p=0.2054
p=0.0567
p=0.9768
p=0.9026
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

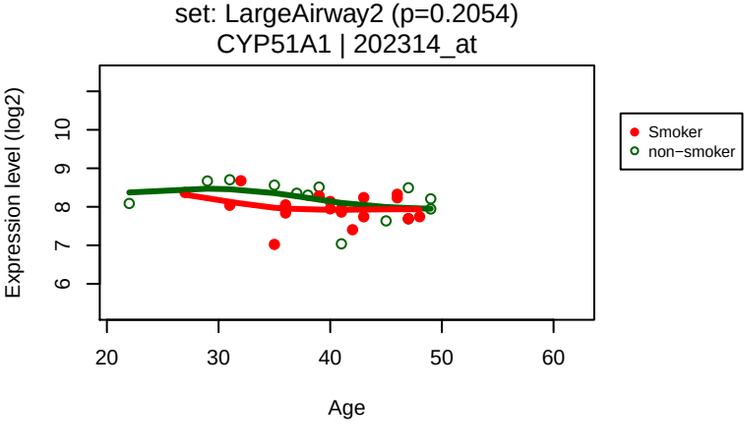

Gene expression ratio (smokers/nonsmokers)
gene: CYP51A1 | 202314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1219
p=0.5862
p=0.7012
p=0.0317
p=0.2054
p=0.0567
p=0.9768
p=0.9026
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

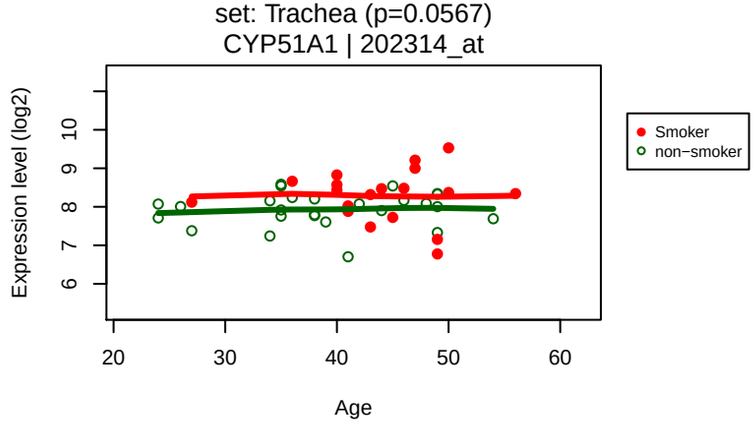

Gene expression ratio in Males
gene: CYP51A1 | 202314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9697
p=0.0255
p=0.2119
p=0.2534
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP51A1 | 202314_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2217
p=0.7378
p=0.3018
p=0.1484
SmallAirway2
LargeAirway2
Trachea
Macrophage2

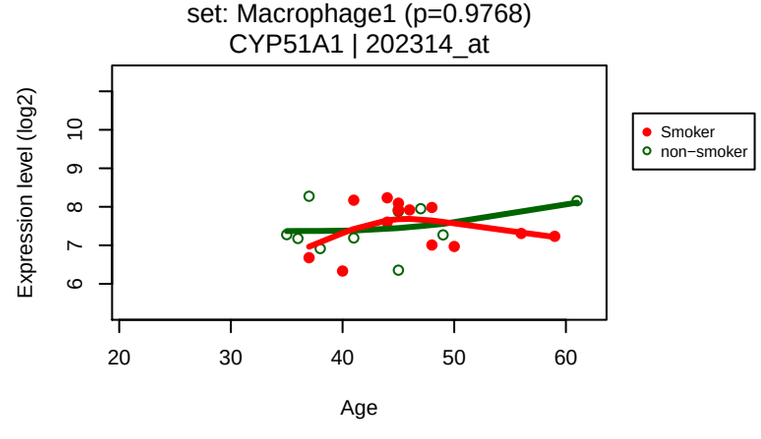

set: SmallAirway1 (p=0.1219)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker

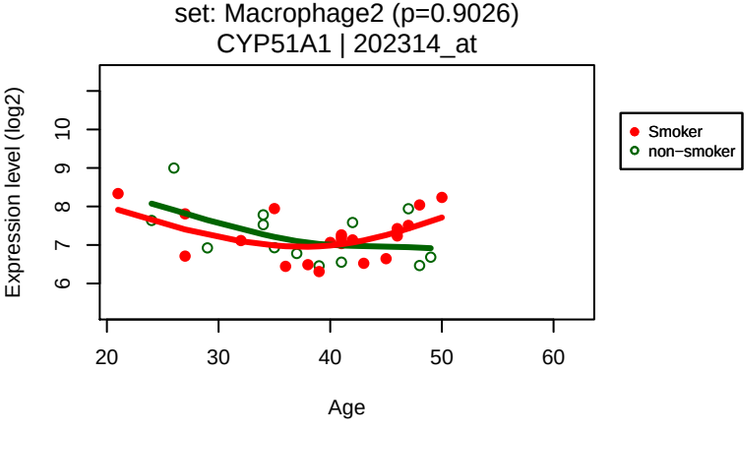

set: SmallAirway2 (p=0.5862)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker

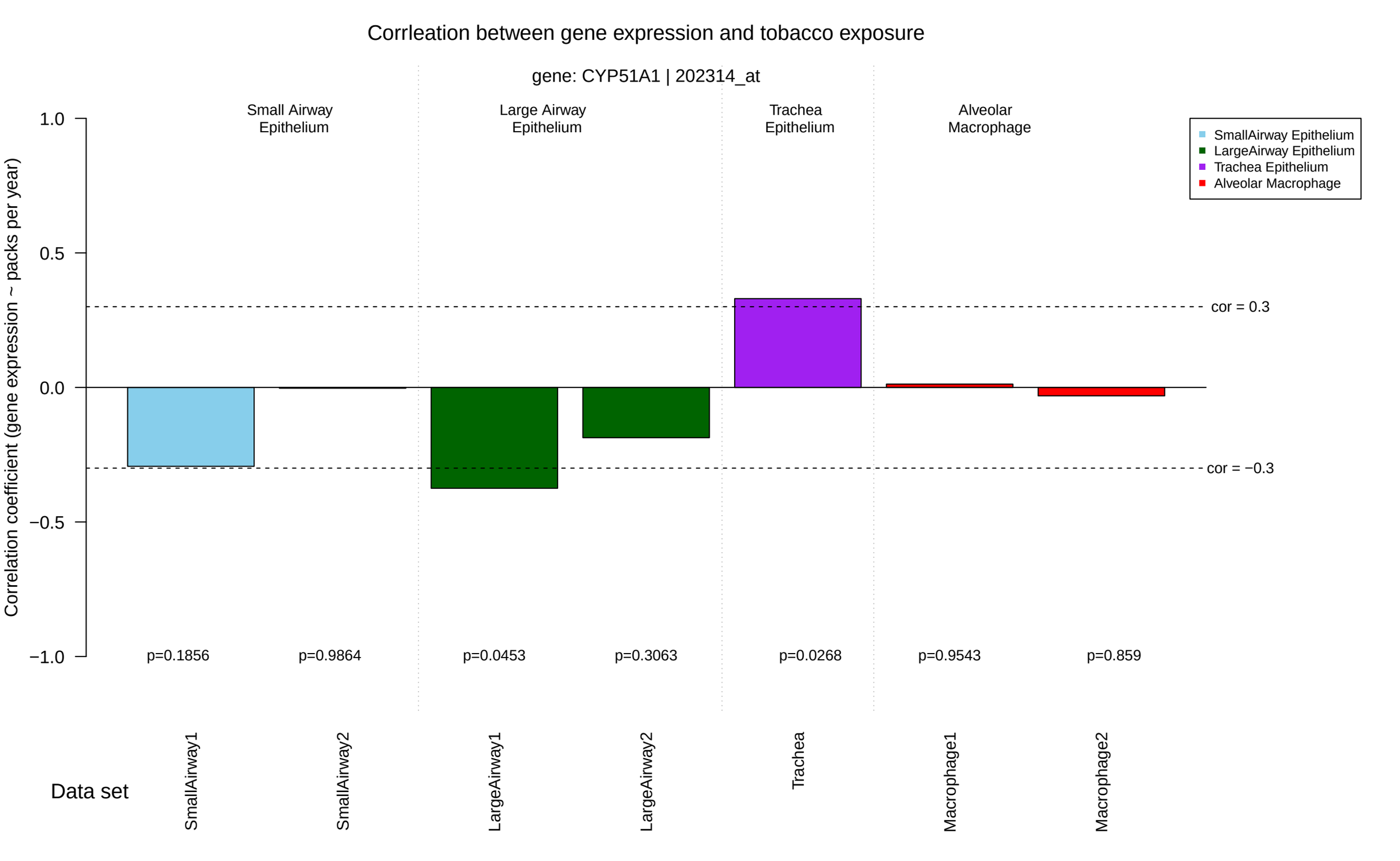

set: LargeAirway1 (p=0.0317)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2054)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0567)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9768)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9026)
CYP51A1 | 202314_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP51A1 | 202314_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1856
p=0.9864
p=0.0453
p=0.3063
p=0.0268
p=0.9543
p=0.859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2